MINING DEALER BEST PRACTICE

A Lean Strategy for Reducing R&I Downtime through Planning and Implementation for Major Components

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

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1.0 Introduction

PCR (Planned Component Replacement) is one of the most important events during the life cycle of a major component because it can have significant impact of productivity on the site: large financial investment, large machine downtime and large amount of manpower.

Customers usually have the challenge of balancing the need to do the PCR with the mine's production requirements. Any gains related to PCR's can have a positive impact on the TCO of the assets as well as the overall customer perceptions of the equipment and support provided.

A Brazilian customer was having trouble achieving this balance, they challenged Sotreq to optimize the R&I (Removal & Install) of major components so that they could meet the production at the same time completing all of the required PCR's. The immediate focus was the 797F fleet's C175-20 LRC engine as the R&I for this component was causing significant delays in achieving production. Sotreq, based on history and their M&R (Maintenance & Repair) knowledge and experience, quickly understood that the R&I downtime contribution suffered from many factors related to resources – parts & component, manpower, tooling, etc.

Sotreq set about optimizing the R&I of the 797F C175-20 LRC Engine through lean strategic planning / preparation and implementation.

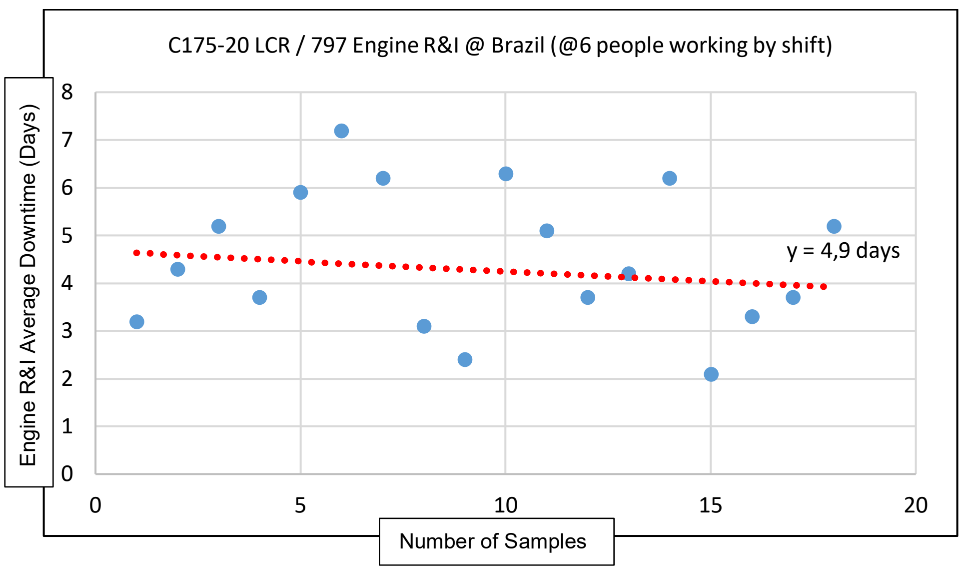

Figure 1 – (Spot Example: 18 samples) Some Historical of R&I downtime for C175-20 LRC in 797 fleet @Brazil (Planned + Unplanned situations), Average of Engine R&I = 4,9 days

2.0 Best Practice Description

The Problem Solving® methodology was used by a multi-functional experienced team (mechanics, component planned, technical support, etc.) looking to identify the main "waste" contributors during the planning and execution of the C175-20 LRC R&I (Removal & Install). The key waste contributors were tooling logistics, tooling availability, handling/lifting capabilities and manpower as shown Figure 1.

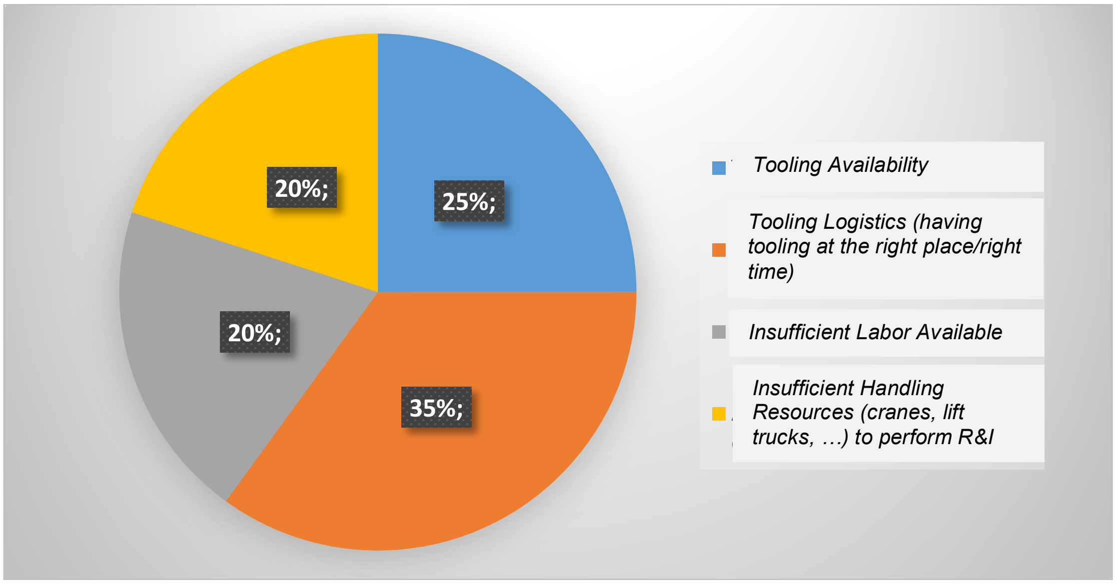

Figure 2 – Waste in C175-20 R&I in Copper Site @Brazil

This group also evaluated the implementation of improvements that did not depend on the actual stoppage of the equipment. The preparation/configuration (set up) of the stock/exchange C175-20 LRC engine, and the development of work with consideration to safety and risk mitigation were incorporated into the R&I process with the following activities:

- Installation of peripheral engine parts (pre-lube, starting motor, Oil Renewal System –(ORS))
- Installation of the torque convertor onto the stock/exchange engine
- Installation of the intake hose on the stock/exchange engine
- Pre-assembly of hoses and clamps for faster installation
- Adherence to contamination control
- Definition and organization of the R&I layout (area, tooling, etc.) to execute the tasks
- Definition of roles, responsibilities, work tasks within the R&I process

The key to success for this work was to anticipate and standardizes of the activities necessary for the R&I. Best Practice 1.01-210827-1 provides a good demonstration of this process.

Another important success factor was high involvement/engagement of the team through understanding individual task as well as the overall objective of the R&I. Throughout the R&I process from planning to execution, meetings were held maintain the focus on and the partnership to achieve the R&I goal.

3.0 Implementation Steps

This Best Practice is focused on the C175-20 engine, but the basic strategy and steps outlined below could be applied to essentially any PCR / major component R&I. Sotreq believes other Dealers and Customers can increase equipment availability and reduce downtime through the implementation of this best practice.

Step 1: R&I Planning & Preparation: Plan for the R&I of the C175-20 engine before stopping the equipment by defining the MACRO details involved:

1. Tools needed for the R&I execution
2. Gant Chart/chronogram detailing all activities and the time (sequence and duration) for the manpower
3. Activities associated with the configuration of the stock/exchange engine
4. Step-by-step documentation (aligned with the OMM's)
5. Meetings/discussions to accompany the R&I process
6. Work area layout

- **Tools**
  Tool availability directly impacts the time and quality of the execution of the service. Understanding this we developed the list based on OMM reference, the experience of the execution team and the review of other best practice literature.

  Tools are separated and put on standby one day before the scheduled R&I in order to minimize the impact of unavailability of the machine and inefficient use of manpower to procure the tooling throughout the shop. All actions that improve efficiency and/or reduce non-productive time should be sought out, considered, and planned to support an efficient and effective equipment stop.

TOOLING LIST - R&I C175	
Tooling	QTY
Electrical Impact 1/2"	3
Pneumatic Impact 1/2"	2
Pneumatic Impact 1"	1
Torque Wranch 1"	1
Torque Wranch 1/2"	1
Hand ratchet 1/2"	1
Drive Extension short 1/2"	1
Drive Extension long 1/2"	1
Socket 36 x 1"	1
Socket 19mm	1
Socket 11mm long	4
Socket 18mm	4
Socket 16mm	4
Combination Wrench 18mm	4
Combination Wrench 16mm	4
Combination Wrench 13/16"	1
Combination Wrench 21mm	1
Combination Wrench 22mm	1
Combination Wrench 24mm	1
Combination Wrench 1 1/8"	1
Combination Wrench 1 3/16"	1
Combination Wrench 1 5/8"	1
Combination Wrench 1 3/8"	1
Talha de 3 ton	2
Lift table	1
Oil collector	3
Wipes	3
Metal tape	2
Lifting Sling 10 ton	2
Lifting Sling 3 ton	1
Hydraulic Lift 100 ton	1

Figure 3 – Tooling List required to Perform an Optimized 797 Engine R&I

- **Gant Chart / Chronogram**

It is essential to have a detailed plan/schedule with resources, activities, and durations. For this project, the plan was developed based on the removal and installation processes in the machine OMM along with the modifications based on the technical team's experience and the customer requirements (i.e., maintenance targets, production targets, etc.).

As the process was based on lean modeling, the R&I was divided into three key work areas: 1) Engine/Radiator Front, 2) Engine Sides and 3) Rear Engine and Converter.

The team was divided to optimize the work - avoid conflicting work and safety risks.

Figure 4a illustrates the distribution of activities. This chart is objective and is an easy way to know what needs to be done and monitor the progress in order to adhere to the target completion.

Activities	Position on Truck	Equipe	Duration (Hours)	% Done
1. Engine Configuration	-		10:20:00	100%
1.1 Install small components	-	-	2:30:00	100%
Pneumatic Start Motor	-	-	1:00:00	100%
Pneumatic compressior	-	-	0:30:00	100%
Auxiliar water pump	-	-	0:30:00	100%
Pre lubrication motor	-	-	0:30:00	100%
Install prelubrication lines	-	-	0:30:00	100%
Configuration pneumatic start motor	-	-	0:20:00	100%
Install and configuration prelubrication relay	-	-	0:30:00	100%
Configuration pnematic compressior	-	-	0:20:00	100%
Install anti-cavitation lines	-	-	1:00:00	100%
Install inlet hose	-	-	1:00:00	100%
Install water pump hose	-	-	0:20:00	100%
Install priming pump harness	-	-	0:10:00	100%
Install ground cable	-	-	0:10:00	100%
Install Torque Converter	-	-	2:00:00	100%
Install front Engine Mount	-	-	1:30:00	100%
2. Separation of Resources and Toolings				
Manual Tooling (Electrical Wrench, Torque Wrench, ...)	-			100%
Lifiting Material	-			100%
Hydarulic Lift Jack	-			0%
3. REMOVE AND INSTALL ENGINE, TORQUE CONVERTER AND RADIATOR			57:40:00	91%
3.1 Preparation			5:05:00	100%
Washer Truck	-	Equipe 1 / 2	2:00:00	100%
Position truck on bay	-	Equipe 1 / 2	0:30:00	100%
Raise up body	-	Equipe 1 / 2	0:20:00	100%
Install wheel chocks	-	Equipe 1 / 2	0:15:00	100%
Drain Radiator	Frente do Motor	Equipe 1	0:20:00	100%
Drain Engine	Frente do Motor	Equipe 1	0:20:00	100%
Drain Renew Tank	LD do Motor	Equipe 1	0:20:00	100%
Drain Torque Converter	Conversor torque	Equipe 2	0:20:00	100%
Drains Air Tank	Frontal Motor	Equipe 2	0:10:00	100%
Turn Off Supression System	Apoio	Apoio	0:15:00	100%
Remove Telemetric System	Apoio	Apoio	0:15:00	100%

Figure 4a – Master Plan to Perform a LEAN 797 Engine R&I

- **Activities associated with the configuration of the stock/exchange engine**

  Based on the premise of executing the maximum number of activities before the machine is even stopped, the configuration of the stock/exchange engine needs to be done in the weeks preceding the stop. Below are some examples of the activities done to the engine:

  - Installation of the peripheral motor parts (pre-lube, starting motor, Oil Renewal System – (ORS))
  - Installation of the torque convertor on the stock/exchange engine
  - Installation of the intake hose on the stock/exchange engine
  - Configuration of the Engine Pads
  - Routing/Clamping the harnesses for the priming and prelube pumps
  - Fill the Engine Oil

    This list should serve as an example not a comprehensive list. The key to this step is the do as much work as possible to prepare this stock/exchange engine before its installation.

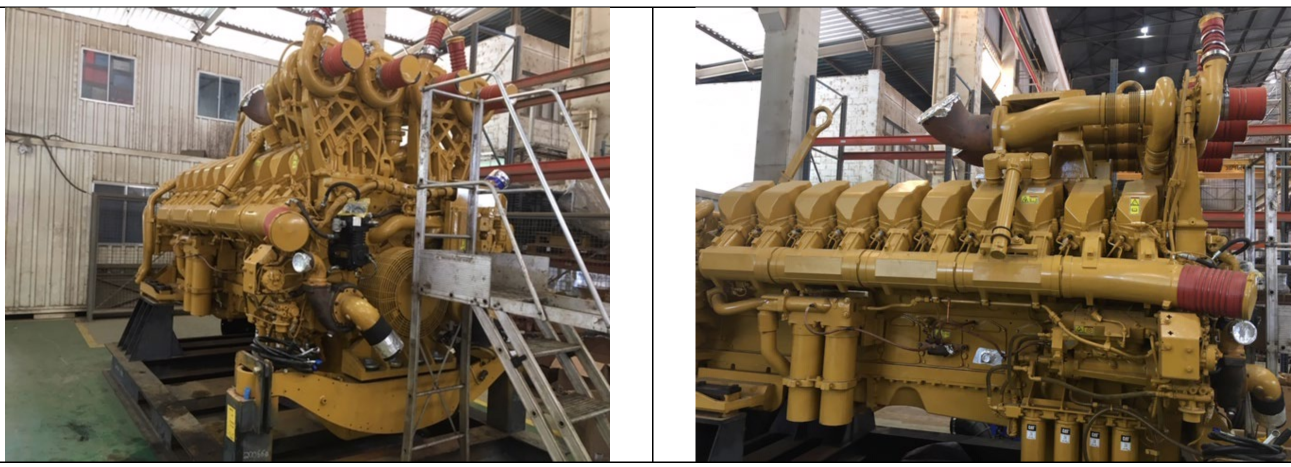

Figure 4b – Spare Engine Set up (part of preparation before machine downtime)

- **Step-by Step Documentation (aligned with the OMM's)**

  To support the activities, all the literature for executing the service/activities of the OMM along with any local additions should be printed out for use on the day of the activity. This material is organized to facilitate efficient consultation according to the division of labor. The following books were printed for the C175-20 LRC R&I:

  Book 01: Radiator Removal Literature
  Book 02: Engine Removal Literature
  Book 03: Engine Installation Literature
  Book 04: Radiator Installation Literature

Figure 5 – Standard Job (documented) by Technician by Process

- **Meetings/Discussions to Accompany R&I**

  A routine of a brief meeting every 30 minutes with the leadership team was established where the plan was updated based on actions completed and alignment was made for the next steps.

  The figure below illustrates the monitoring of the key tasks for both the configuration of the stock/exchange engine and the engine R&I. In order to optimize and facilitate the reading/comprehension of the information by the entire team, the plan was prepared in a simple and objective manner.

  During the entire process, the alignment with the executions team was fundamental and was carried out during the group exchange dialogues and objective meetings where the work objectives, defined strategy and work progress were reinforced. Team input and alignment early in the process were essential. Team involvement made the team more committed and focused on the end result. Reinforcing all information and efficiency gains must be considered from the planning to the execution of the activities.

Resources			
Activies to Configuration Engine			
Activies	**Responsible**	**Status**	**Observations**
Kit to install Engine	Marcio	Complet	Material was delivery 11th Dec
Kit to install Torque Converter	Marcio	Complet	
Components			
Pneumatic Compressor	Marcio	Complet	Pedido emitido, previsão de entrega 11/12
Auxiliary water pump	Marcio	Complet	
Start Pneumatic Motor	Marcio	Complet	
Renew Valve	Marcio	Complet	
Engine Prelubrication pump	Marcio	Complet	
Position the Engine on Bay	Aparecido	Complet	
Lubrication Engine	Aparecido		

Activies to R&I Engine			
Activies	**Responsible**	**Status**	**Observations**
Hydraulic Jack Lift	Marcio		
Prepare lift material	Carlos		Provide resources one day before
Prepare manual tooling	Carlos		Provide resources one day before
Provide Electrical Impact Wrench	Ediclei		Provide resources one day before
Provide Pneumatic Wrenchs	Ediclei		Provide resources one day before
Provide Oil Collector	Ediclei		Provide resources one day before
Provide Clean ELC Collector	Ediclei		
Provide Stairs	Ediclei	Complet	Using the resoruces for Preventive BOX
Dormentes (Calçar motor e radiador)	Wallace		Provide resources one day before
Provide electrical to rework harness	Fábio	Complet	
Programação da equipe de AFEX e Despacho	Marcio		

Figure 6 – Follow up Dashboard of Management team

- **Work Area Layout**

As a reminder, the entire process is based on the lean philosophy focused on reducing waste in the process and optimizing end time by having all resources where they are needed when they are needed.

After careful analysis of the resources and activities a “two box work area” layout was proposed, validated by the site security team, and implemented for this site. The equipment was positioned perpendicular to the overhead crane as shown in Figure 7 below.

Each dealer/customer will need to determine the appropriate layout for their site.

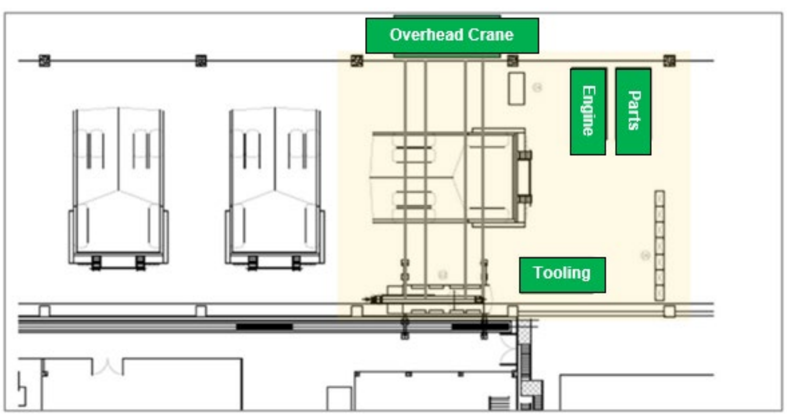

Figure 7 – Standard LEAN Layout (Particular for this Case Study)

It should also be noted that the approximately 80 hours of manpower were required for preparation and configuration before the machine was even stopped for this engine R&I.

Step 2: R&I Implementation:

The R&I Planning and Preparation Task results in the following:

- ✓ LEAN Plan in Place – A detailed GANT chart is in place based on customer requirements, manpower and other resource availability.
- ✓ R&I Bay Ready – All Parts & Consumables, Tooling, Documentation, Fluids are properly placed in the work bay.
- ✓ Replacement Engine Ready – A staged component is ready for immediate installation.

The truck should be taken from production, washed, positioned in the work bay at the appropriate time when all the above three items are in place.

The key R&I Implementation Tasks are as shown in Figure 8.

Key R&I Tasks
Drain Fluids
Deconfigure & Remove Radiator
Remove Center Bar
Deconfigure & Remove Engine & Torque Converter
Configure Radiator
Install Engine, Torque Converter & Radiator
Configure Electrical, Hydraulic, Inlet & Exhaust Systems
Implement Torque Procedures
Add Fluids and Lubrication
Implement Test Procedures
Implement Liberation/Release Procedures.

Figure 8. Key R&I Tasks – Please use appropriate documentation for the specific component.

Activities are grouped by position on the equipment and a logical sequence. Like a Formula 1 team, each mechanic is engaged in their own specific set of critical tasks contributing to the optimized R&I. All tooling, parts and other resources are available and appropriately placed in the workbay contributing to the optimized R&I.

Note: In addition to a Gant chart with the entire plan in the maintenance book, a white board is in use in the bay with key timing info (arrival, estimated departure, etc.), personnel, additional backlogs/work and more; Sotreq is in the considering implementing a monitor to track the status of the work in real time.

there is an opportunity to improve this process, that is to put into the workshop, a monitor display to help the team drives the task to achieve the downtime target. We did not implement it on this case.

4.0 Benefits

Key Benefits:

- Implementation of the Lean Strategy described in this best practice focused on 1) R&I Planning & Preparation and 2) R&I Implementation will increase equipment availability through less downtime. The planning/preparation is key to success as the machine should not be taken down until all the resources (including the replacement component) are in place and ready to be used in the actual R&I.

Sotreq Results from this Specific Case:

The results from the implementations were widely acknowledged by the customer showing that Sotreq was able to meet the company's needs in minimizing actual downtime to meet production targets at this particular time but also that Sotreq has a good planning / preparation strategy that could be applied for all R&I's.

Sotreq would be happy to show other dealers the specifics of the plan, preparation, resource requirements for this specific case; please feel free to contact one of the authors.

5.0 Resources Required

Key Resources include the following:

- Planning and Engineering Personnel
- Execution Team Personnel
  - Mechanics – to execute the activities according to the lean model
  - Project Leader – to monitor overall effective and efficient completion of work / address needs
  - Supervisor(s) - to monitor mechanics and areas of work / address needs
- Tooling – required for each specific activity (based on OMM and local work guidelines)

  Note: For additional details please reach out to Sotreq.

6.0 Supporting Attachments / References

DMAIC Methodology, 6 Sigma Process; Lean Manufacturing Guidebook, VPS – Customer Production System Guideline, 794AC Service Manual.

7.0 Related Best Practices

Best Practice (1.01-210827-1) – A PM Model to Deliver a Higher Availability for 794AC Trucks

8.0 Acknowledgements

This Best Practice was written by:

Júlio César Corrêa
Manager, Product Performance and Engineering
Julio.correa@sotreq.com.br
+ 55 94 98111-6441

Fábio Cota
Product Performance Consultant (R&I Project Engineer)
fabio.cota@sotreq.com.br
+ 55 94 98807-8155

In Colaboration with Product Support team of SOTREQ in Salobo Mine, Maintenance Team of Customer, and **Douglas Friedrich, Site Performance Manager of CATERPILLAR**.

Marcio Nunes Griffith
797F Maintenance Supervisor of Customer

Carlos Travassos
Product Support Coordinator
carlos.travassos@sotreq.com.br
+ 55 94 981230515

Leonardo Rezende
Product Support Manager
Leonardo.rezende@sotreq.com.br
+ 55 94 9255-6644